

3. Theoretical Exercise: Properties of Bootstrap

We will now derive the probability that a given observation is part of a bootstrap sample.

Let $Z = \{Z_1, \dots, Z_n\}$ be our original sample of size n . (For example, $Z_i = (X_i, Y_i)$ can represent the pair of predictors and target variables for each observation $i = 1, \dots, n$.)

Let $Z^* = \{Z_1^*, \dots, Z_n^*\}$ be a bootstrap sample of size n , resampled from Z . (You can think of it as one of the B bootstrap samples $Z^{*1}, \dots, Z^{*B}$ used for bagging.)

We want to show that, for each $i = 1, \dots, n$, the probability that Z_i is part of the bootstrap sample is $P(Z_i \in Z^*) = 1 - 1/e$.

- What is the probability that the first bootstrap observation is not the i -th observation from the original sample? Justify your answer.
- What is the probability that the second bootstrap observation is not the i -th observation from the original sample?
- Argue that the probability that the i -th observation is not in the bootstrap sample is $(1 - \frac{1}{n})^n$.
- When $n = 5$, what is the probability that the i -th observation is in the bootstrap sample?
- When $n = 100$, what is the probability that the i -th observation is in the bootstrap sample?
- When $n \rightarrow \infty$, what is the probability that the i -th observation is in the bootstrap sample?

$$a) P(Z_1^* \neq Z_i) = 1 - P(Z_1^* = Z_i) = 1 - \frac{1}{n}$$

$$b) P(Z_2^* \neq Z_i) = 1 - P(Z_2^* = Z_i) = 1 - \frac{1}{n}$$

$$\vdots$$

$$f) P(Z_n^* \neq Z_i) = 1 - \frac{1}{n}$$

$$c) P(Z_i \notin Z^*) = P(Z_1^* \neq Z_i, \dots, Z_n^* \neq Z_i) = P(Z_1^* \neq Z_i) \cdot \dots \cdot P(Z_n^* \neq Z_i)$$

$$= \left(1 - \frac{1}{n}\right) \cdot \dots \cdot \left(1 - \frac{1}{n}\right) = \left(1 - \frac{1}{n}\right)^n$$

$$d) n=5 \quad P(Z_i \in Z^*) = 1 - P(Z_i \notin Z^*) = 1 - \left(1 - \frac{1}{n}\right)^n = 1 - \left(1 - \frac{1}{5}\right)^5 = 0.672$$

$$e) n=100 \quad P(Z_i \in Z^*) = \dots = 1 - \left(1 - \frac{1}{100}\right)^{100} = 0.634$$

$$f) n=\infty \quad P(Z_i \in Z^*) \xrightarrow{\text{numerically (see)}} 0.632 = 1 - \frac{1}{e}$$

More formally:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = \exp(x)$$

$\downarrow x = -1$

$$P(Z_i \in Z^*) = 1 - \left(1 + \frac{(-1)}{n}\right)^n \xrightarrow{n \rightarrow \infty} 1 - e^{-1} = 1 - \frac{1}{e}$$

$\Rightarrow$ On average, 63.2% of original data Z is in the bootstrap sample Z^*

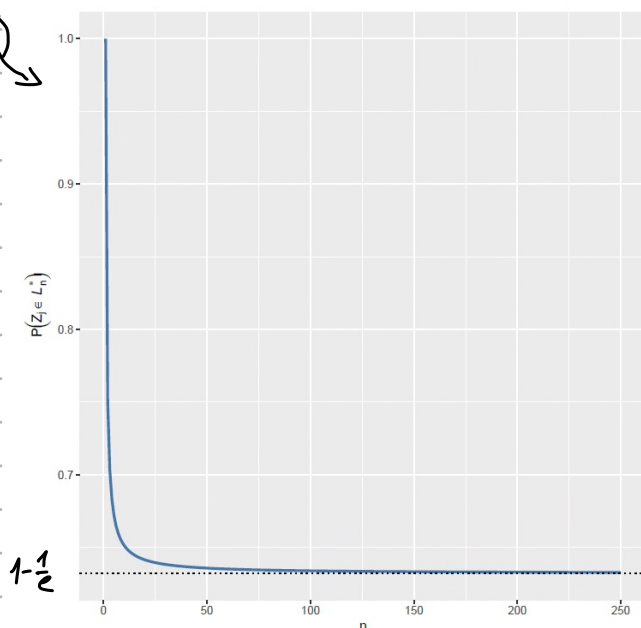


Figure 4: Probability that Z_i is in the bootstrap sample of size n .

$\leadsto$ The remaining $\sim 36.8\%$ of observations can be used for "out-of-bag" test error estimates in bagging and Random Forests